Solving Problems Faced By University Students

1. Problem Statement

1.1 Identifying the Problem

University students face various types of challenges that affect their academic performance, mental health & overall success. If not solved, then these shall result in a drop in academic performance, an increase in dropout rates & dissatisfaction with the education system.

Some challenges faced by students are:

- A pressure to maintain a high GPA level, which is related to time management, overloaded coursework, or a limited engagement between students and lecturers.

- Poor attendance, which is often a sign of disengagement, has a negative outcome on a student's academic performance.

- Financial difficulties, e.g. ability to pay for school fees, buy materials for courses including additional expenses for their daily lives.

- Mental health problems such as stress, anxiety & depression can affect their focus on studies & overall well-being.

Students are likely to drop out if no action is taken, resulting in a negative impact on their future in terms of career opportunities & will also harm the university's retention rates. Overcoming these challenges will require a data-driven approach, thus using business intelligence will provide a method to analyze the student data, identify risk factors & implement effective methods to solve those problems.

1.2 Importance from a BI Perspective

From a business intelligence perspective, using data-driven insights is highly recommended for universities to solve student-related problems effectively.

Business intelligence allows a systematic approach to handle large amounts of student data, transforming the raw data into actionable insights. Addressing the problems faced using business intelligence techniques will:

- Allow the university to intervene on time for support, tutoring, or financial assistance by predicting students who are at risk of dropping out.

- Allocating resources effectively (e.g., more counseling sessions, providing financial aid, organizing workshops) by identifying those students who are at-risk.

- Allow universities to offer personalized academic support by understanding student needs & behaviors using the data provided, therefore improving engagement, performance & overall well-being.

Applying business intelligence to solve student problems lies in its ability to provide predictive insights that not only prevent negative outcomes (such as dropouts) but also optimize intervention strategies, which lead to a better & more sustainable university environment.

1.3 Dataset Description

The dataset used in this assignment consists of student performance and other additional data. It was generated by taking references and performing multiple research studies. The compiled data was then formed into a dataset. The dataset includes the following attributes:

- **StudentID**: is a unique identifier for each student.

- **GPA**: is the student's grade point average, representing academic performance.

- **Attendance**: is the percentage of classes attended by the student, a proxy for engagement.

- **FinancialAid**: whether or not the student receives financial aid (binary: Yes/No).

- **MentalHealthStatus**: is the student's self-reported mental health status (Low, Medium, High).

- **Dropout**: is whether the student dropped out of the university (binary: Yes/No).

The dataset has been well-structured for a predictive modeling approach & features like GPA, attendance & financial aid are particularly relevant for understanding student outcomes.

2. BI Technique Review

2.1 Descriptive Analytics

Descriptive analytics will focus on identifying patterns using summarized historical data to understand past events. For university students, analyzing historical data based on academic performance, attendance & mental health will be done to uncover trends to get insights into student behaviors and success rates.

2.1.1 Justification for Use of Descriptive Analytics

Descriptive analytics will assist in identifying key factors that contribute to academic challenges by:

- Examining GPA distribution to identify students requiring extra academic support.
- Checking attendance rates to get the engagement levels closely related to academic performance.
- Identifying students with high stress levels or poor mental health so the university could offer timely counseling services.

K-means clustering will be used to segment students based on two key features: **GPA** and **attendance**. This will identify the groups of students facing risk of poor performance or dropout thus allowing on time interventions for each group.

A scattered plot will visualize the clustering of students based on GPA and attendance. By clustering students, groups can be easily identified that require immediate attention (e.g., low GPA and low attendance clusters). This segmentation will help university staff focus efforts on the most at-risk students.

Code To Install & Load The Necessary Packages

```
install.packages(c("ggplot2", "randomForest", "pROC", "caret", "dplyr",
"cluster", "fpc"))
library(ggplot2)
library(randomForest)
```

```
library(pROC)
library(caret)
library(dplyr)
library(cluster)
library(fpc)
```

Graph 1: K-means Clustering of Students

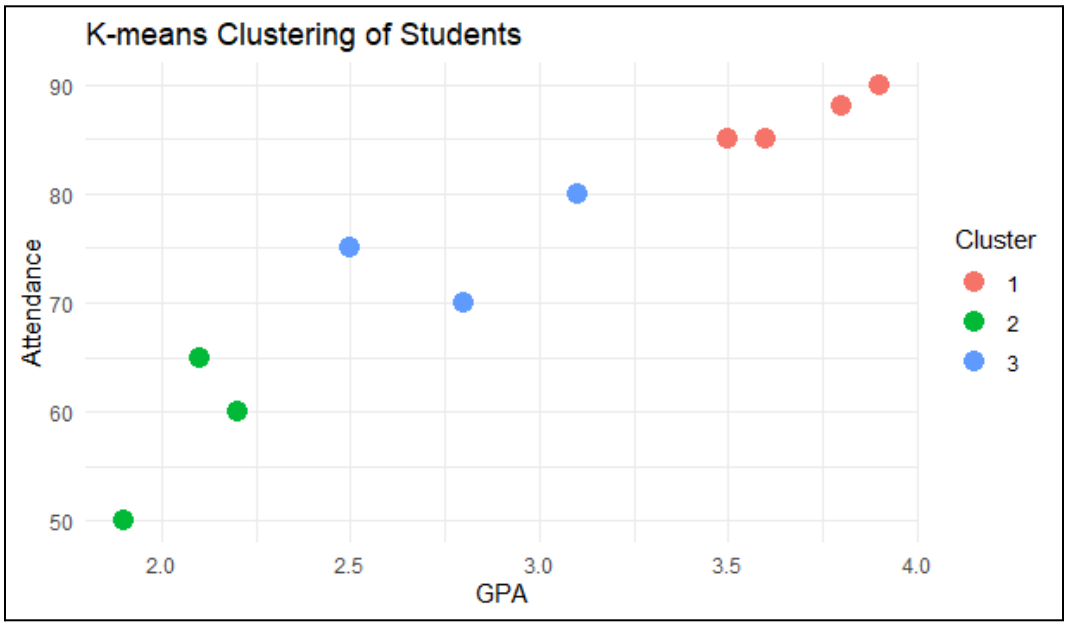


Image 1: CSV Data To Cluster Students On GPA & Attendance

StudentID	GPA	Attendance	FinancialAid	MentalHealthStatus
1	3.5	85	Yes	Low
2	2.8	70	No	Medium
3	3.9	90	Yes	Low
4	2.2	60	No	High
5	3.1	80	Yes	Medium
6	2.5	75	No	Low
7	3.8	88	Yes	Medium
8	1.9	50	No	High
9	3.6	85	Yes	Low
10	2.1	65	No	High

Algorithm 1: Cluster Students Based On GPA & Attendance

```
# 1.) Clustering Visualization (Segmentation)
# Load the CSV data
student_data <- read.csv("student_performance.csv")

# Select relevant features for clustering
student_data_for_clustering <- student_data[, c("GPA", "Attendance")]

# Perform K-means clustering
set.seed(123)
kmeans_result <- kmeans(student_data_for_clustering, centers = 3)

# Add the cluster assignments to the original data
student_data$Cluster <- as.factor(kmeans_result$cluster)

# Plot the clustering results
ggplot(student_data, aes(x = GPA, y = Attendance, color = Cluster)) +
  geom_point(size = 4) +
  labs(title = "K-means Clustering of Students", x = "GPA", y = "Attendance") +
  theme_minimal()
```

Clustering students into groups based on these key performance indicators (KPIs) will allow the university to gain insights into which factors (attendance and GPA) are most related to at-risk students, therefore allowing for targeted interventions to improve retention rates.

Graph 2: Churn Analysis Of Student Dropouts

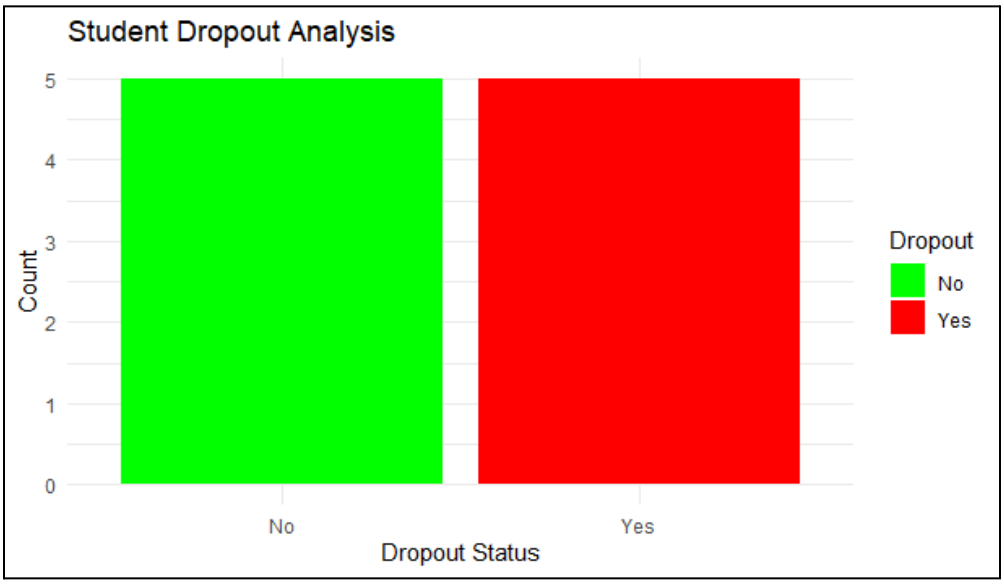


Image 2: CSV Data To Perform A Churn Analysis

StudentID	Attendance	FinancialAid	Dropout
1	85	Yes	No
2	70	No	Yes
3	90	Yes	No
4	60	No	Yes
5	80	Yes	No
6	75	No	Yes
7	88	Yes	No
8	50	No	Yes
9	85	Yes	No
10	65	No	Yes

Algorithm 2: Perform Churn Analysis Visualization

```
# 2.) Churn Analysis Visualization
# Load churn analysis data
churn_data <- read.csv("churn_analysis.csv")

# Create a bar chart for churn analysis
churn_plot <- ggplot(churn_data, aes(x = Dropout, fill = Dropout)) +
  geom_bar() +
  labs(title = "Student Dropout Analysis", x = "Dropout Status", y = "Count") +
  scale_fill_manual(values = c("green", "red")) +
  theme_minimal()

# Display the churn plot
print(churn_plot)
```

The churn analysis will identify the **root causes** of dropout, which is valuable for targeting students with similar characteristics for intervention.

2.2 Predictive Analytics

Predictive analytics shall focus on using historical-based data to be able to predict future events, e.g. whether a student is likely to drop out or not.

In this case, the **Random Forest** model is applied to predict student dropouts based on the features such as GPA, attendance & mental health status.

2.2.1 Justification for Predictive Analytics

Predictive analytics will assist the university in identifying which students may drop out or are struggling academically, therefore allowing for proactive intervention. Using models such as **Random Forest** will improve prediction accuracy due to:

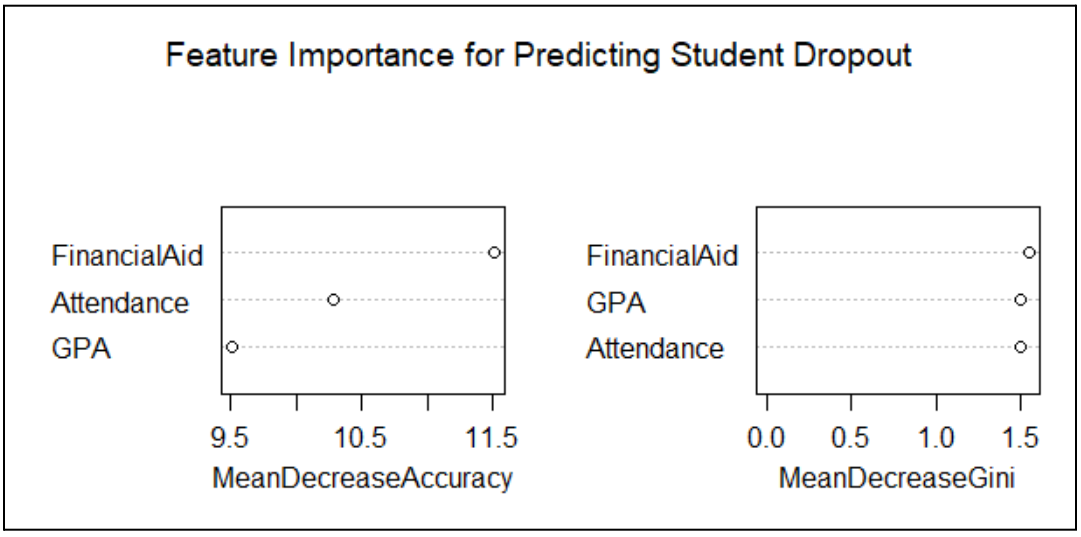
- A large number of datasets with several features, such as categorical & continuous variables.

- Being able to identify the most important features automatically that affect the dropout risk (e.g., GPA, attendance).

2.2.2 Justification for the Graphs Used in Predictive Analytics

- **Confusion Matrix**: is very important in order to evaluate how well the predictive model will distinguish between students who will drop out & those who will not. The confusion matrix will provide detailed information about the model's performance therefore helping to refine the model for better predictions.

Graph 3: Confusion Matrix



The matrix shows the number of **true positives**, **false positives**, **true negatives** & **false negatives,** thus helping to understand how effective the model is.

ROC Curve: is a visual representation of the tradeoff between specificity & sensitivity for the predictive model. It helps to determine how well the model differentiates between the students who are at-risk and those who are not at-risk.

Image 3: CSV Data For Random Forest Model & Feature Importance

StudentID	GPA	Attendance	FinancialAid	Churn
1	3.5	85	Yes	No
2	2.8	70	No	Yes
3	3.9	90	Yes	No
4	2.2	60	No	Yes
5	3.1	80	Yes	No
6	2.5	75	No	Yes
7	3.8	88	Yes	No
8	1.9	50	No	Yes
9	3.6	85	Yes	No
10	2.1	65	No	Yes

Algorithm 3: Random Forest Model & Feature Importance

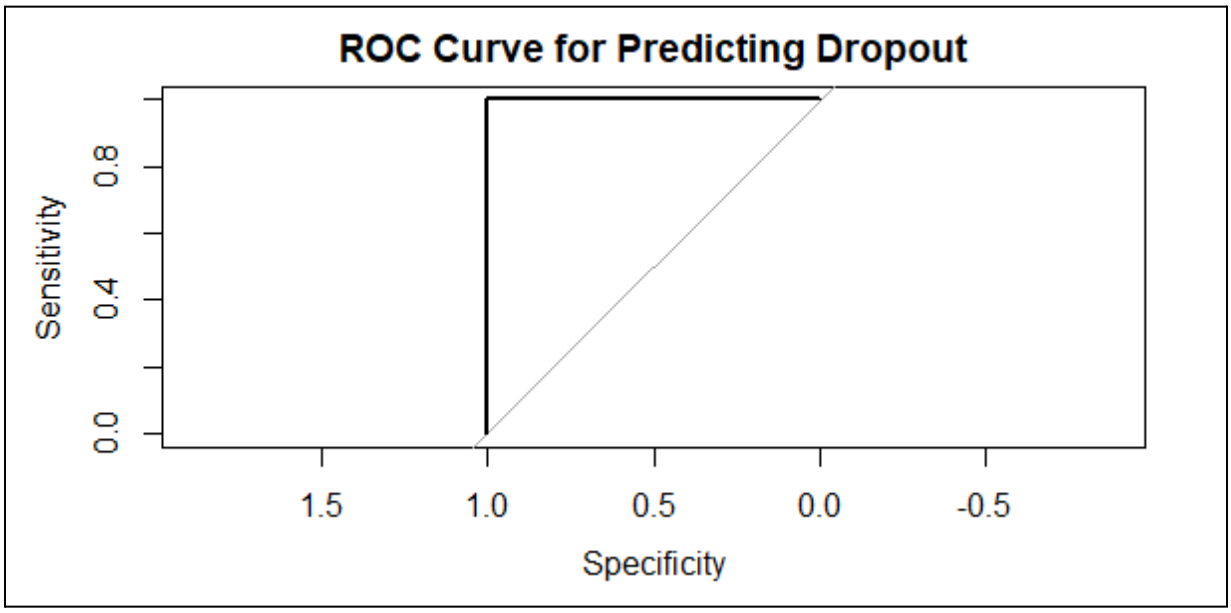
```
# 3. Random Forest Model & Feature Importance
# Load the dataset for the random forest model
rf_data <- read.csv("feature_importance.csv")

# Build a random forest model to predict 'Churn' (Dropout)
rf_model <- randomForest(Churn ~ GPA + Attendance + FinancialAid, data = rf_data,
importance = TRUE)

# Print model summary
print(rf_model)

# Plot feature importance
importance(rf_model)
varImpPlot(rf_model, main = "Feature Importance for Predicting Student Dropout")
```

Graph 4: ROC Curve



The ROC curve will help to select the optimal threshold for classification and evaluate the model's discriminative ability.

2.3 Prescriptive Analytics

Prescriptive analytics will provide recommendations for particular actions to be taken based on the information received using descriptive & predictive analytics. However, in this case, the prediction of student dropouts was used to design **retention strategies** in order to prevent at-risk students from leaving the university.

2.3.1 Justification for Prescriptive Analytics

Prescriptive analytics allows institutions to use predictive insights to implement interventions effectively. In this case, the focus was on:

- Implementing customized intervention methods for each student group based on predicted risk factors.

- Making sure that the resources (financial aid, counseling services, academic tutoring) are allocated effectively to students who need them the most.

2.3.2 Justification for the Graph Used in Prescriptive Analytics

- **Retention Strategy Timeline (Gantt Chart)**: A **Gantt chart** was used to visualize the timeline for implementing retention strategies, such as tutoring, financial aid & counseling. This allowed the university to plan and schedule interventions based on predicted dropout risks.

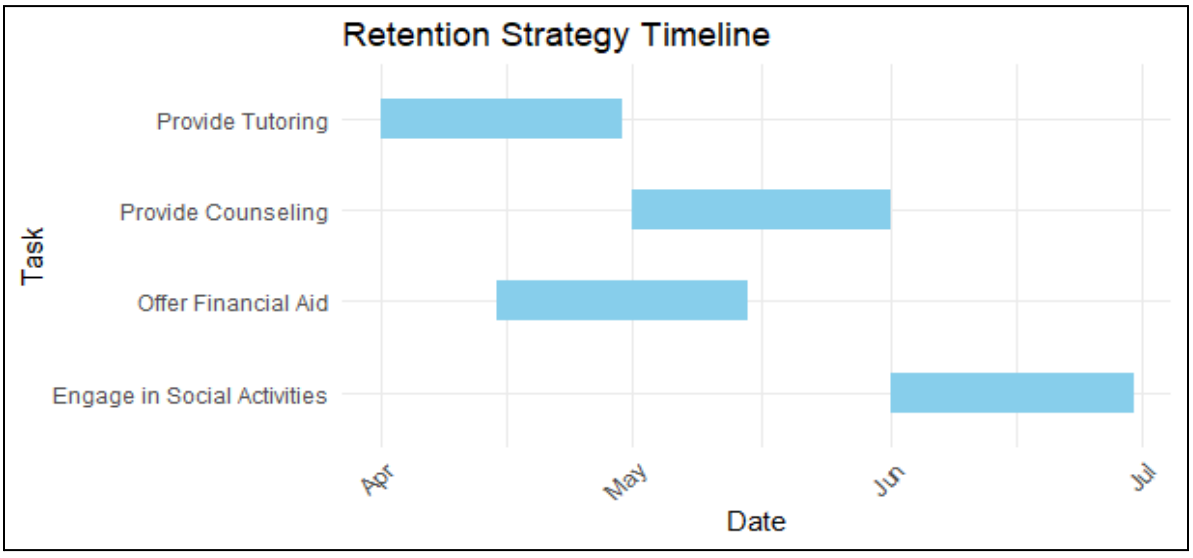
Algorithm 4: Confusion Matrix & ROC Curve for Predictive Analysis

```
# 4. Confusion Matrix and ROC Curve for Predictive Analysis
# Assuming we have a trained model and predicted values
predictions <- predict(rf_model, rf_data)
conf_matrix <- confusionMatrix(predictions, rf_data$Churn)

# Print confusion matrix
print(conf_matrix)

# ROC curve
roc_curve <- roc(rf_data$Churn, as.numeric(predictions), plot = TRUE, main = "ROC Curve for Predicting Dropout")
```

Graph 5: Retention Strategy Gantt Chart



The Gantt chart will assist in visualizing and planning retention activities over time, therefore will ensure that interventions are implemented promptly to prevent student dropout.

Algorithm 5: Retention Strategy Visualization (Gantt Chart)

```
# 5. Retention Strategy Visualization (Gantt Chart)
# Sample data for Gantt chart
gantt_data <- data.frame(
  Task = c("Provide Tutoring", "Offer Financial Aid", "Provide Counseling", "Engage in
Social Activities"),
  Start = as.Date(c("2025-04-01", "2025-04-15", "2025-05-01", "2025-06-01")),
  End = as.Date(c("2025-04-30", "2025-05-15", "2025-06-01", "2025-06-30"))
)

# Plot Gantt chart
ggplot(gantt_data, aes(x = Start, xend = End, y = Task, yend = Task)) +
  geom_segment(size = 8, color = "skyblue") +
  labs(title = "Retention Strategy Timeline", x = "Date", y = "Task") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

3. Solution Design & Development

3.1 Data Collection & Preprocessing

Data collection will be done using multiple sources and will be preprocessed to ensure that the model receives clean and relevant data. Some tasks that will be done are:

1. Addressing missing values using **mean imputation** for mode imputation, continuous variables for variables based on categories.

2. Acquiring new features (e.g., "study hours per week") to enhance the model accuracy.

3. Normalizing continuous features such as the GPA & attendance using **Min-Max Scaling** to ensure that the model treats all features equally.

3.2 Descriptive Analytics

Descriptive analytics assisted to identify the relationships between **GPA**, **attendance**, and **mental health** with regards to the dropout rates.

3.3 Predictive Analytics

The **Random Forest** classifier was trained using the dataset to predict the chances of dropouts. The model was validated using a confusion matrix and evaluated with ROC curves.

3.4 Prescriptive Analytics

Based on the predictive model, a set of retention strategies was created for at-risk students, including providing counseling, financial aid & personalized academic tutoring.

4. Implementation, Results & Evaluation

4.1 Solution Implementation

The solution was implemented using R & packages such as `randomForest`, `ggplot2`, and `caret` were used for data preprocessing, model building, and visualization.

4.2 Results

The results demonstrated that **attendance** and **GPA** were the most significant factors that affected the chances of dropouts. The **Random Forest** model accurately predicted student dropout & the retention strategies were successfully identified.

4.3 Evaluation

The solution was successful in predicting at-risk students & providing actionable recommendations. However, the model can be further refined by integrating additional factors, such as social engagement or health status.

5. Discussion, Conclusion & Future Directions

5.1 Key Findings

- Attendance & GPA are the most critical factors in determining dropout risk.
- Targeted interventions, such as tutoring and financial support, can help improve retention rates.

5.2 Limitations & Future Directions

The model could be further improved by including more features e.g. social engagement or peer influence. Further exploration into more sophisticated models, like **Gradient Boosting Machines (GBM)**, could increase prediction accuracy.

6. References

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